GRAVITY
ELEVATING SOLUTIONS

USER MANUAL

Gravity Vertical System

GVS Vertical Lifeline Fall-Arrest System — Safe Use, Inspection & Limitations



Document No.	GG-GVS-UM
Revision	Rev. 6
Applicable standard	EN 353-1 (Personal fall protection equipment. Guided type fall arresters including a rigid anchor line)
Intended users	Trained & authorized climbers

Read and understand this manual before using the system. Keep it available to all users.

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Contents

1. Introduction 3
1.1 About the GVS 3
1.2 About this manual 3
1.3 Key terms 3
1.4 Intended use 4

2. Certification 5

3. Preliminary Checks 5
3.1 User requirements 5
3.2 Personal protective equipment (PPE) 6
3.3 System inspection 6

4. Use of the System 8
4.1 Before climbing 8
4.2 Ascending the vertical lifeline 10
4.3 Descending the vertical lifeline 10
4.4 Disconnecting from the lifeline 11
4.5 Common mistakes to avoid 11

5. Limits of Use 11

6. Prohibited Use 12

7. In the Event of a Fall 13

8. Inspection, Maintenance & Storage 13
8.1 Inspection schedule 13
8.2 Maintenance and storage 13

9. Component Summary 14

1. INTRODUCTION

1.1 About the GVS

The Gravity Vertical System (GVS) is a vertical fall-arrest system that provides continuous fall protection for authorized users climbing fixed ladders and similar permanent access structures.

The system consists of a vertical steel lifeline secured to the structure by top and bottom anchor assemblies. Intermediate brackets are installed at intervals of less than 10 meters to support the lifeline, improve system stability and reduce the loads transferred to the structure during a fall.

When used correctly, and together with approved personal protective equipment (PPE), the GVS helps to prevent serious injury or death resulting from a fall from height.

1.2 About this manual

This manual explains the safe use, inspection, limitations and maintenance of the GVS only. **All users must read and understand this manual before using the system.** It should be kept at the site and made available to every user.

> **i NOTE**
> This user manual excludes all other Permanent Vertical Lifelines.

Symbols used in this manual

Symbol	Meaning
⚠ **WARNING**	A hazard that could result in serious injury or death if the instruction is not followed.
i NOTE	Important information that supports correct and safe use.
✓ **MANDATORY**	A check or action that must be completed before use.

1.3 Key terms

Term	Definition
Vertical Lifeline	The vertical steel cable along which the glider travels.
Glider (fall arrester)	A device that travels along the lifeline and automatically locks in the event of a fall. Use a glider **without a shock absorber** on systems fitted with an **inline shock absorber**, and a glider **with a shock absorber** on systems **without** an inline shock absorber (See Component Summary)
Captive glider	A glider that remains permanently on the lifeline and is stored in a glider box when not in use.

Term	Definition
Removable glider	A glider that the user attaches to the lifeline before climbing and removes afterwards.
Competent person	A person trained and experienced enough to identify hazards and defects in the system and authorized to take corrective action.
Fall clearance	The minimum free space required below the user to arrest a fall without striking the ground or an obstruction.

1.4 Intended use

The GVS is intended **exclusively** for use as a personal fall-arrest system on fixed ladders and permanent access structures. The system is designed to:

- Provide continuous fall protection during ascent and descent.
- Minimize fall distance by engaging automatically in the event of a fall.
- Allow users to pass intermediate brackets without disconnecting, where compatible equipment is installed.

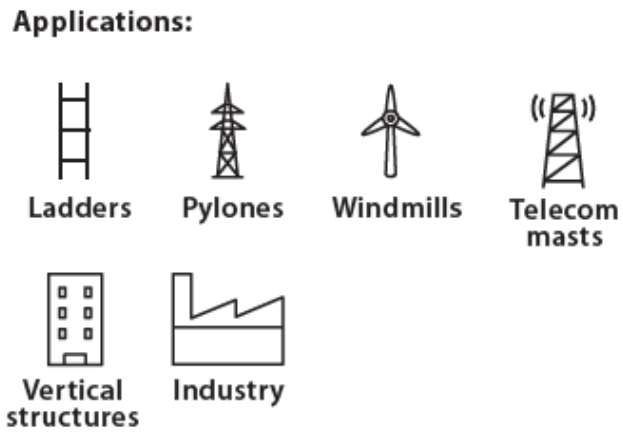


Figure 1 – Typical applications of the GVS

> **NOTE**
>
> The GVS is **not** designed for:
>
> - Work positioning.
> - Rope access activities.
> - Material lifting.
> - Suspension work.
> - Any application not specifically approved by Gravity Vertical Systems.

2. CERTIFICATION

The Gravity Vertical System has been designed and tested in accordance with the following standards:

Standard	Scope
EN 353-1:2014+A1:2017	Guided type fall arresters including a rigid anchor line

Only approved components supplied or specified by Gravity Vertical Systems may be used with the system.

> ⚠ WARNING
>
> Only use GVS compatible Equipment. Substituting components may compromise system performance and will invalidate certification & may lead to serious injury, property damage, and death

3. PRELIMINARY CHECKS

Before each use, the user shall verify that the system, personal protective equipment and work environment are suitable for safe operation. **The system shall not be used if any defect, damage, missing component or unsafe condition is identified.**

> ✓ IT IS MANDATORY TO CHECK THE FOLLOWING BEFORE USE
>
> All checks in sections 3.1 to 3.3 must be completed before every use of the system.

3.1 User requirements

Before using the GVS, users shall ensure that:

- They are medically fit to work at height.
- They are not affected by vertigo, dizziness, fatigue, alcohol, medication or any condition that may impair safe use of the system.
- They are in possession of a valid Fall Arrest training certificate.
- They are familiar with the operation of the GVS and the associated rescue procedures.
- Their body weight is **at least 60 kg**, and their total weight including clothing, tools and equipment **does not exceed 150 kg**.
- A minimum of **two competent persons** are present in the work area so that emergency assistance can be provided if required & must have valid Fall Arrest & Basic Rescue training and kit on site that is ready to use.

> ⚠ WARNING
>
> **Working alone while using the GVS is prohibited.**

3.2 Personal protective equipment (PPE)

The following minimum PPE is required when using the GVS:

- A full-body harness with a front (sternal) fall-arrest attachment point.
- A helmet with a chin strap.
- Suitable safety footwear and gloves.



Figure 2 – Full-body harness

Before use:

1. Inspect the harness for damage, wear, contamination or unauthorized modification.
2. Verify that all buckles and adjustment points function correctly.
3. Ensure the harness fits correctly and is securely adjusted.
4. Inspect the helmet for cracks, deformation, UV degradation, damaged suspension or missing components.
5. Verify that the helmet chin strap is correctly fastened and adjusted before climbing.
6. Do buddy checks.
7. Confirm that all PPE inspections are current.

> ⚠ WARNING
>
> A correctly fitted climbing helmet with a secure chin strap shall be worn at all times while climbing, working at height or using the GVS.
>
> An incorrectly fitted harness or damaged PPE may result in serious injury or death during fall arrest.

3.3 System inspection

Before climbing, verify that:

1. The annual inspection is current – check the inspection date on the system identification label (Figure 3).
2. There is no visible damage, deformation, corrosion or excessive wear.
3. The cable is correctly tensioned – check the tension indicator (Figure 4).
4. Intermediate brackets are secure and undamaged.
5. Top and bottom anchor assemblies are secure.
6. The glider moves freely and operates correctly (function test done at ground level).
7. Adequate fall clearance exists below the work area.
8. A site-specific rescue plan is available and can be implemented immediately when required.

9. Rescue equipment is immediately accessible & ready to use.

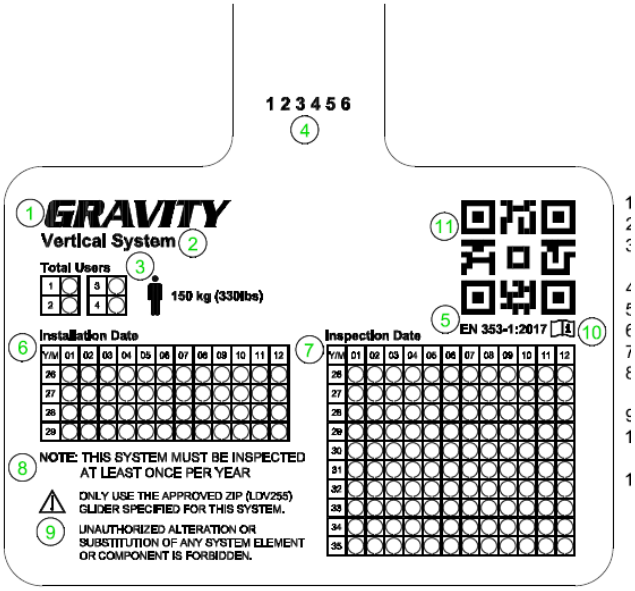


Figure 3 – System identification and inspection label



Figure 4 – Tension indicator: incorrect (top), correct (bottom)

Figure 3 – label field key

No.	Label field	No.	Label field
1	Manufacturer details	7	Inspection date
2	System type	8	System must be inspected at least once per year
3	Maximum number of users and maximum weight	9	Only the approved glider may be used
4	Serial number	10	User manual must be read before use
5	System standard	11	QR code – inspection certificate and user manual
6	Installation date		

⚠ WARNING

If any defect or abnormal condition is identified, remove the system from service immediately and report it to a competent person.

Repairs and component replacement shall only be carried out by a certified Gravity Vertical Systems installer.

4. USE OF THE SYSTEM

4.1 Before climbing

Complete the following steps in order before starting to climb.

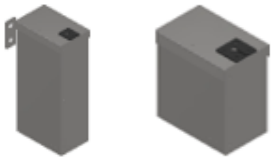
1 **Inspect your full-body harness.**
Carry out the PPE checks in section 3.2.

2 **Inspect the vertical lifeline system.**
Check the cable, brackets, anchors and glider as described in section 3.3.

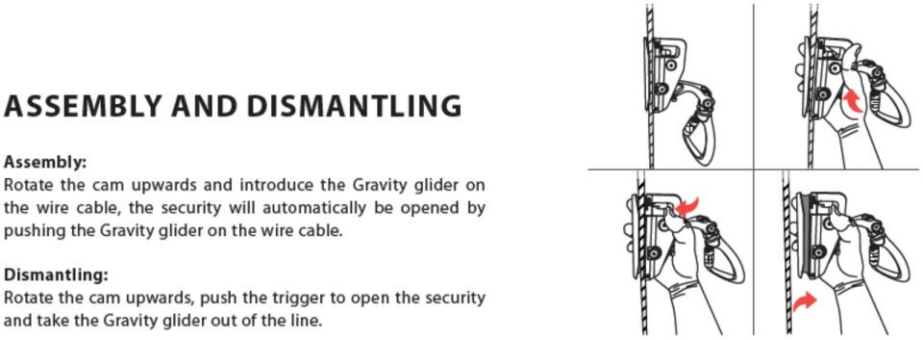
3 **Check the tension indicator.**
Confirm the tension indicator shows the correct (green) condition – see Figure 4.

4 **Confirm the system inspection is current.**
Check the inspection date on the system label and make sure there is no visible damage.

5 **Access the glider.**
Captive glider: open the glider box and remove the glider.



Removable glider: rotate the cam upwards and place the glider onto the wire cable; the security catch opens automatically when the glider is pushed onto the cable.



6

Verify the glider orientation.

Make sure the glider is fitted the right way up on the cable, according to the arrow/marking on the glider. A glider fitted upside down will not arrest a fall.





7

Connect to your harness.

Connect the glider to the **front fall-arrest attachment point** on your harness using the approved connector. Check that the connector gate is fully closed and locked.



8

Test before you climb.

While standing on the ground or a safe platform, pull the glider connector sharply downward to confirm that the glider locks on the cable, then release it.

4.2 Ascending the vertical lifeline

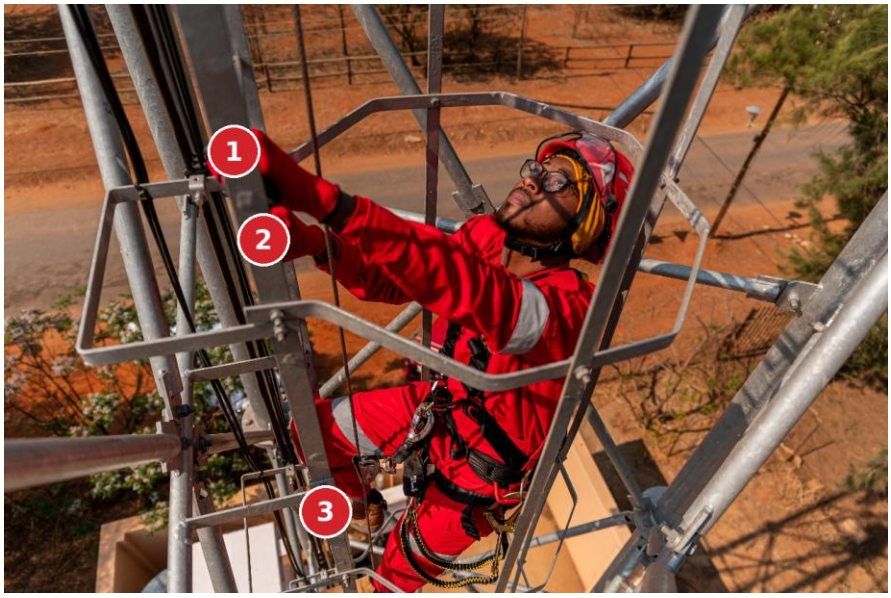


Figure 5 – Maintain three points of contact (1–3) while climbing

- Maintain three points of contact with the ladder where possible.
- Climb normally without pulling or forcing the glider.
- Allow the glider to follow your movement automatically.
- Pass intermediate brackets without disconnecting from the system (on systems designed for this purpose).
- Keep your body centered on the ladder and avoid excessive leaning.

4.3 Descending the vertical lifeline

- Face the ladder and descend in a controlled manner.
- Allow the glider to travel down the cable naturally as you move.
- Do not hold the glider open, override the locking mechanism or pull the glider downward.
- Maintain secure footing and three points of contact where possible.
- Disconnect only once you have reached ground level or a safe landing platform, in accordance with site procedures.

4.4 Disconnecting from the lifeline

1. Make sure you are in a safe area, or connected to another fall-protection device, before disconnecting.
2. Remove the glider connector from the front attachment point on your harness.
3. **Captive glider:** return the glider to the glider box and close it.
4. **Removable glider:** rotate the cam upwards, push the trigger to open the security catch and remove the glider from the cable (Figure 6).

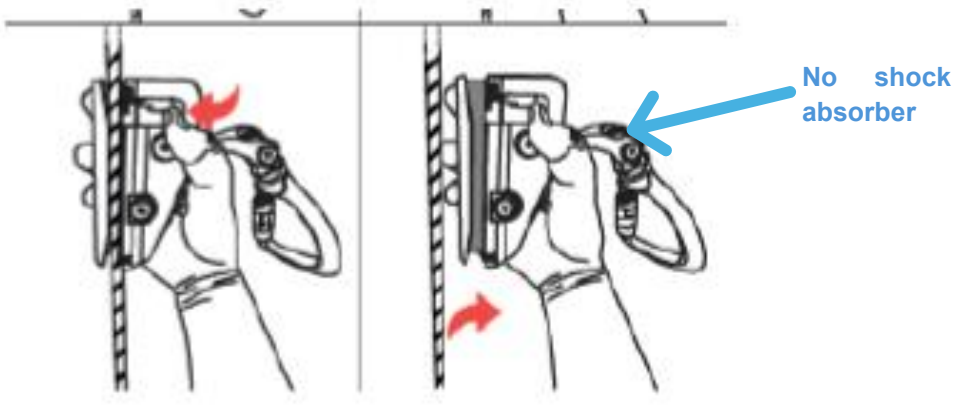


Figure 6 – Removing a removable glider from the lifeline

4.5 Common mistakes to avoid

⚠ COMMON **MISTAKES**

- **DO NOT** climb while disconnected from the glider.
- **DO NOT** fit the glider upside down.
- **DO NOT** pull the glider by hand instead of allowing it to track naturally.
- **DO NOT** bypass or tamper with the locking mechanism.
- **DO NOT** carry loads that prevent safe ladder climbing.
- **DO NOT** descend too quickly.

5. LIMITS OF USE

Parameter	Limit
Minimum user weight	60 kg (body weight, excluding tools and equipment)
Maximum user weight	150 kg (including clothing, tools and equipment)
Users per glider	One
Maximum users on the system	As shown on the system identification label
Intermediate bracket spacing	Less than 10 m

Parameter	Limit
Maximum rescue time	15 minutes

- Before removing the glider from the lifeline, or before detaching from the fall-arrest system, fall protection must be provided by another device.
- The lifeline must not be used as a work-positioning system. Where positioning is required, a separate restraint or positioning system must be used.
- Handling the glider while climbing or descending may interfere with the braking mechanism. If necessary, handle the glider **by its connector only** – never hold the body of the glider while in use.
- The system should not be installed in a highly corrosive atmosphere, because of the risk of non-visible stress corrosion, unless specific control measures are in place or compatibility has been established.

6. PROHIBITED USE

The system **must not** be used:

⚠ PROHIBITED

- By a lone worker – a minimum of two competent persons must be present.
- By more than one person per glider. The glider is for a single user, who is responsible for it.
- By more users than the maximum number shown on the system label.
- Where the fall clearance below the user is insufficient.
- If the energy absorber has deployed or has extended by more than 10 cm.
- With any energy absorber other than the one originally fitted to the fall arrester.
- In combination with additional safety equipment that has not been approved by Gravity Vertical Systems, this may be detrimental to safety.
- Where there is a risk of electric shock.
- During frost or risk of frost, thunderstorms or any other dangerous weather conditions.
- As an earth (ground) for welding machines.
- If it has been stored in an environment that is aggressive to its materials.
- If there is any doubt about its condition or safety.
- After arresting a fall, until a competent person has confirmed in writing that it is safe to use.
- For any purpose other than that intended by the manufacturer.

7. IN THE EVENT OF A FALL

A person suspended in a harness after a fall is at risk of **suspension trauma** (orthostatic intolerance), which can become life-threatening within minutes. A prompt rescue is essential.

1. Raise the alarm immediately and activate the site-specific rescue plan.
2. Carry out the Rescue Plan using the site rescue equipment, immediately.
3. While waiting for rescue, the suspended user should, if possible, move their legs and push against the ladder to relieve pressure from the harness leg loops or use suspension trauma strap.
4. Once rescued, the user must receive medical assessment, even if no injuries are visible.
5. Remove the entire system, including the glider, energy absorber and harness, from service. Do not use it again until it has been inspected and confirmed safe in writing by a competent person.
6. Report and record the incident according to site procedures.

8. INSPECTION, MAINTENANCE & STORAGE

8.1 Inspection schedule

Inspection	Frequency	Carried out by
Pre-use check (section 3)	Before every use	User
Periodic detailed inspection	At least once every 12 months (more often in harsh environments)	Competent person / certified Gravity installer
Post-incident inspection	After any fall arrest or suspected damage	Competent person / certified Gravity installer

8.2 Maintenance and storage

- Keep the lifeline, brackets and glider free of dirt, paint, ice and other contamination.
- Clean removable gliders with fresh water and a mild detergent only; allow them to dry naturally, away from direct heat.
- Do not lubricate, modify or dismantle the glider unless instructed by Gravity Vertical Systems.
- Store removable gliders and harnesses in a clean, dry, ventilated place, away from direct sunlight, chemicals and sharp edges.
- Keep captive gliders in the glider box when not in use.

9. COMPONENT SUMMARY

The main components of the Gravity Vertical System are shown below.



Top Anchor Assembly

Anchors the lifeline to the structure and transfers fall loads into it.



Intermediate Bracket

Guides and supports the cable at intervals of less than 10 m, reducing load on the anchors.



Bottom Anchor Assembly

Secures the lower end of the lifeline and maintains cable tension.



Captive Glider

Fall arrester that remains permanently on the lifeline; stored in a glider box.



Removable Glider

User-installed fall arrester that is attached to and removed from the lifeline.



Inline Shock Absorber

Absorbs energy and reduces forces on the user and structure during a fall.

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